

Construction Crack Detection & Structural Health Monitoring

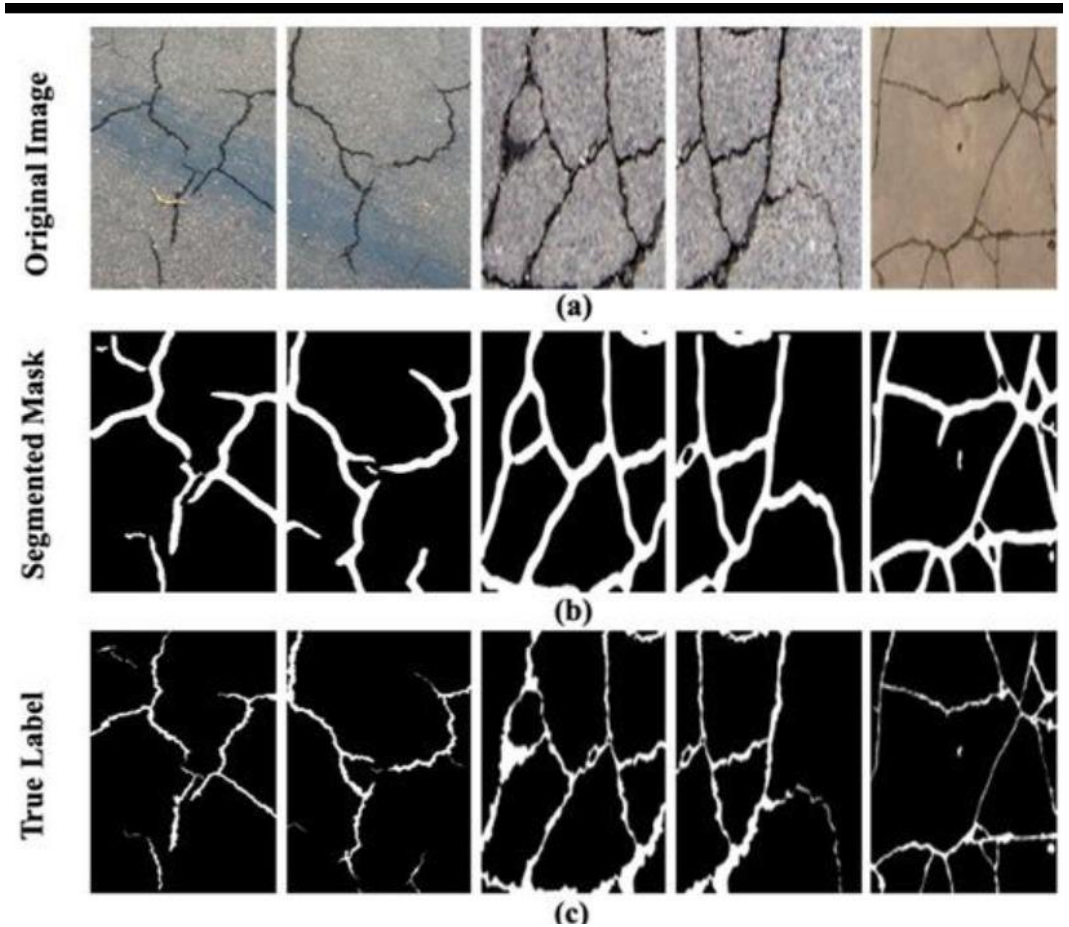
Artificial Neural Network

Submitted By

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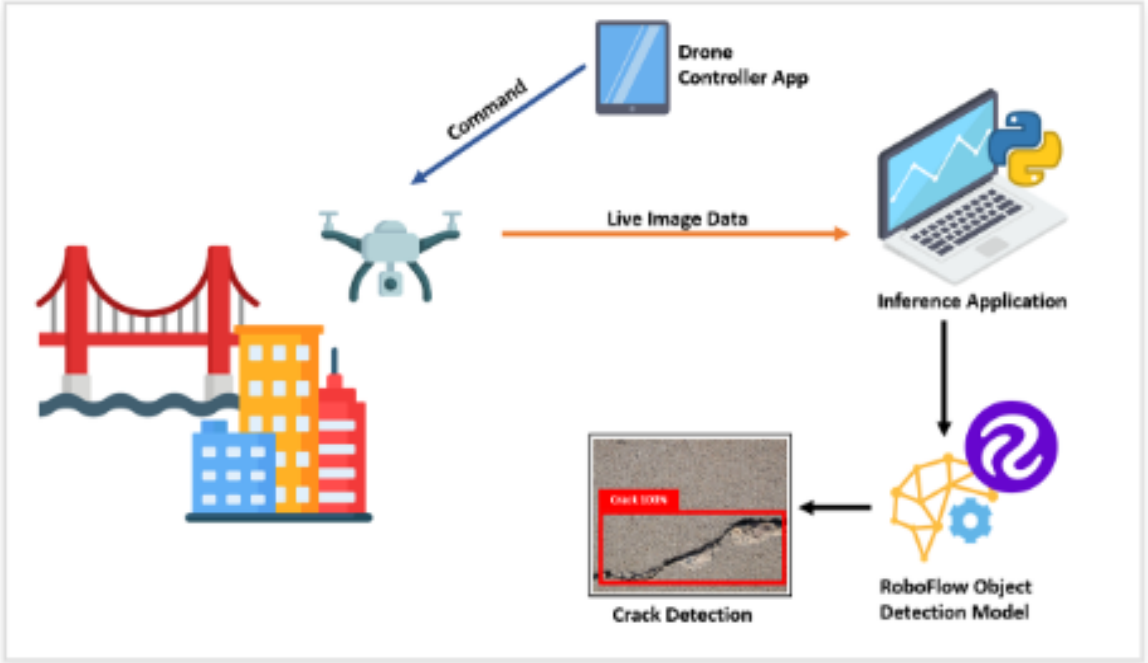
Introduction

- **Goal:** Detect and assess construction cracks from drone/robot camera images to aid structural health monitoring.
- **Approach:** Deep learning pipeline combining CNN classification (ResNet18) and U-Net segmentation for localization.
- **Inputs:** 224×224 .jpg images; drone/UAV and ground-robot captures.
- **Outputs:** Severity labels (intact, slight, moderate, severe) and crack masks/heatmaps for visualization.



Problem Statement

- **Manual inspection challenges:** Time-consuming, costly, and hazardous at scale (bridges, high-rise facades, long spans).
- **Need:** Rapid, objective, and scalable crack assessment using AI on imagery.
- **Objectives:**
  - Detect the presence/absence of cracks
  - Classify severity into intact/slight/moderate/severe
  - Visualize crack density/location via segmentation and heatmaps



Dataset Description

- **Dataset:** severity-dataset for multi-class crack severity classification
- **Image format:** 224×224 resolution, .jpg format only
- **Classes:** Four distinct crack severity classes (intact, slight, moderate, severe)

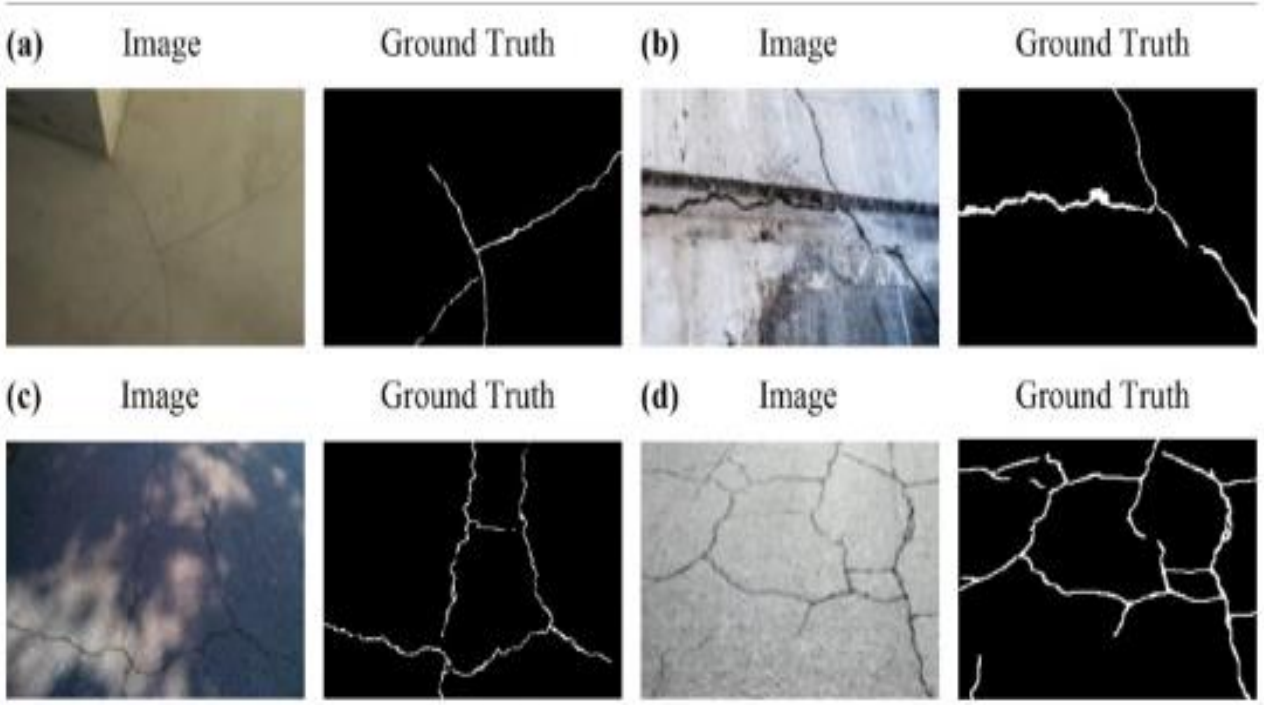
Dataset Split Distribution:

Split	Intact	Slight	Moderate	Severe	Total
Train	1,400	1,400	1,400	1,400	**5,600**
Validation	400	400	400	400	**1,600**
Test	200	200	200	200	**800**
Total per Class	**2,000**	**2,000**	**2,000**	**2,000**	**8,000**

Note: The dataset is perfectly balanced across all classes, which helps prevent model bias during training.

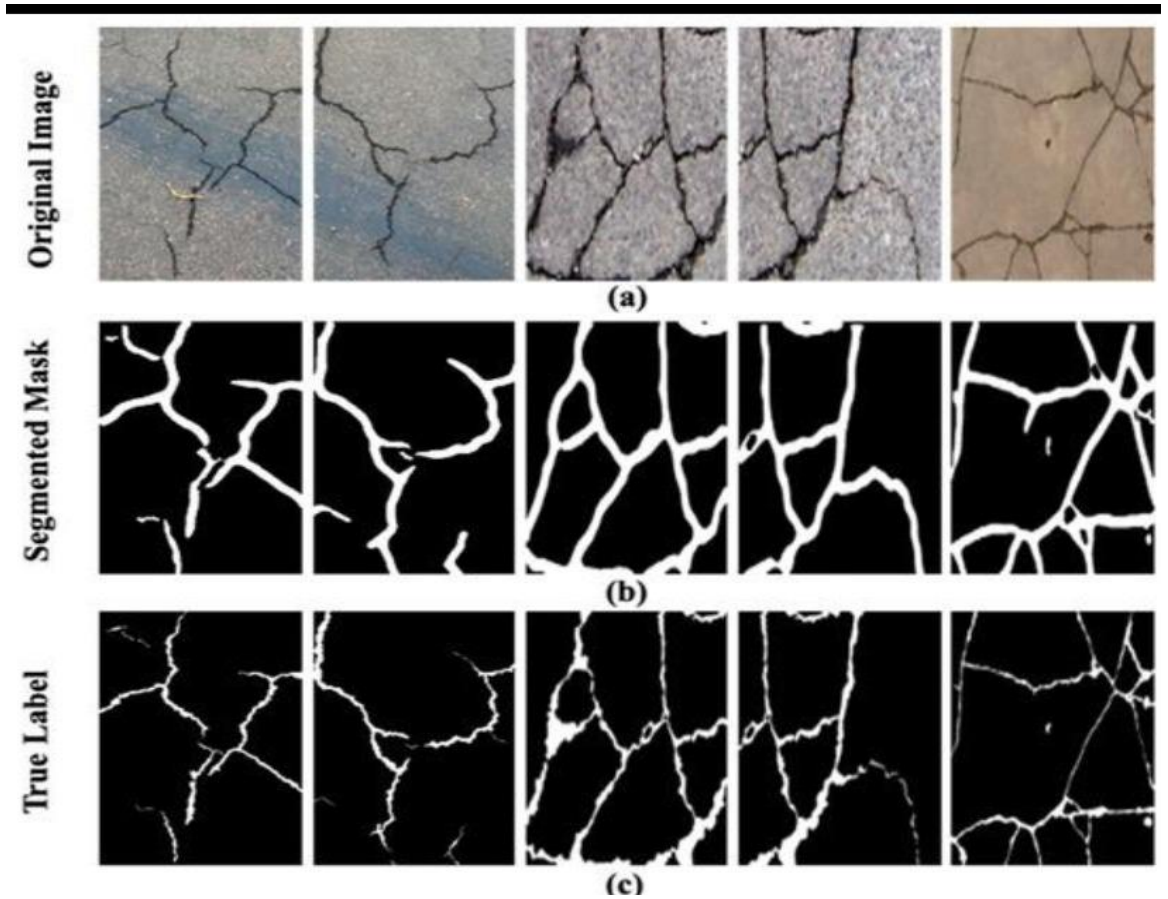
Methodology – Part 1: Crack Severity Classifier (ResNet18)

- **Architecture:** ResNet18 initialized with pre-trained ImageNet weights for feature extraction
- **Modification:** Final fully connected layer replaced with new linear layer outputting 4 classes (intact, slight, moderate, severe)
- **Data Preparation:**
  - **Training:** Resize (224×224), RandomHorizontalFlip, RandomRotation, ColorJitter, ImageNet normalization
  - **Validation/Test:** Resize (224×224), ImageNet normalization
- **Training Parameters:**
  - **Loss:** nn.CrossEntropyLoss for multi-class classification
  - **Optimizer:** Adam with learning rate `1e-4`
  - **Epochs:** 10 **Batch Size:** 32



Methodology - Part 2: Crack Segmentation (U-Net)

- **Purpose:** Binary crack/no-crack segmentation for localization and density visualization
- **Model:** Simplified U-Net (UNetSimple)
- **Encoder:** Stacks of DoubleConv blocks (Conv → BatchNorm → ReLU) with MaxPool downsampling
- **Decoder:** Upsample + skip connections + DoubleConv for feature fusion
- **Output:** 1-channel logits mask for crack presence



Results

Overall Accuracy:

98.38%

- **Class-wise Performance:**

Class	Precision	Recall	F1-score
Intact	0.99	0.98	0.98
Slight	0.98	0.99	0.98
Moderate	0.97	0.98	0.97
Severe	0.99	0.98	0.99

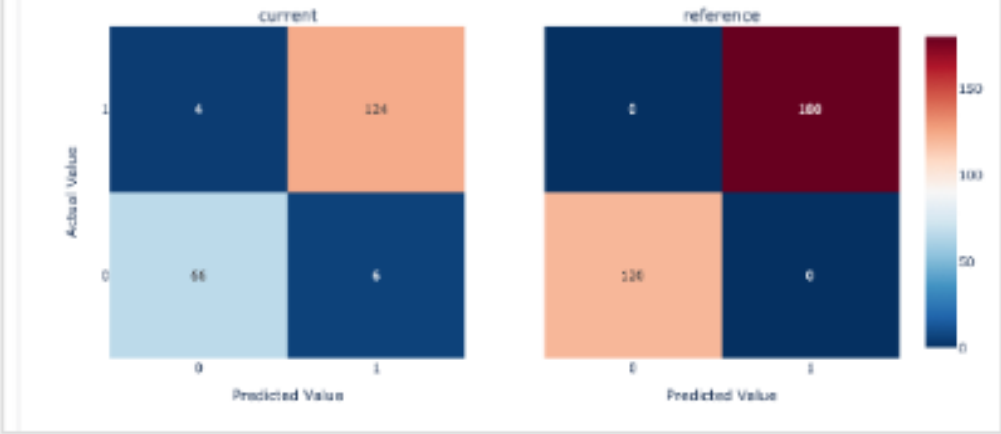
- **Notes:** Balanced dataset and data augmentation contributed to robust generalization across all severity classes.

Confusion Matrix

```
classification_performance_report = Report(metrics=[
    ClassificationPreset(probas_threshold=0.7),
])

classification_performance_report.run(current_data=bcancer_cur, reference_data=bcancer_ref)

classification_performance_report
```



Model correctly classified the vast majority of samples across all classes

Conclusion

Key Achievement:

98.38% Overall Accuracy in Crack Severity Classification

- **Model Performance:** Excellent results across all severity classes (intact, slight, moderate, severe) with balanced precision, recall, and F1-scores.
- **Practical Applications:** Supports safer, faster, and cost-effective structural health monitoring using drones/robots for infrastructure inspection.
- **Deployment Strategy:** Scalable to large infrastructure assets:
  - Integration with drone/robot inspection workflows
  - Prioritization of maintenance based on severity classification
  - Quantitative damage assessment through segmentation maps
- **Future Work:** On-device inference optimization for UAV/UGV platforms, active learning with new field data, integration with structural analysis systems.